

Calculus I for Bio & Social Sciences

Full Course Design

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Course Overview

- **Course name:** Calculus I for Bio & Social Sciences
- **Level:** Undergraduate, introductory
- **Format:** Lecture (3×/week, 50 min) plus one weekly TA/discussion session (50 min)
- **Duration:** 14–15 week semester
- **Assessment weights:** Weekly homework 8%, Quizzes (4) 35%, Group projects (2) 20%, Portfolios (2) 7%, Final exam 30%
- **Assumed background (assumption — not confirmed by instructor):** College algebra / precalculus fluency: function notation, graphing and transforming linear, polynomial, rational, exponential, and logarithmic functions; basic unit-circle trigonometry (sine and cosine, at a recognition level); comfort with algebraic manipulation, factoring, and solving equations. No prior calculus and no programming background are assumed.

1 Module 1: Functions and Mathematical Modeling

Goal: Refresh core function types and show how functions model biological and social phenomena.

1.1 Review of Function Types

Core concepts:

- Definition of a function; domain and range
- Linear functions and constant rate of change
- Polynomial and power functions

- Rational functions and asymptotic behavior
- Transformations: shifts, stretches, reflections

Learning objectives:

- Explain the defining properties of linear, polynomial, and rational functions.
- Apply transformation rules to sketch graphs of shifted or scaled functions.
- Evaluate which function family best matches a given data trend.

1.2 Exponential and Logarithmic Functions

Core concepts:

- Exponential functions and the base e
- Properties of logarithms; the inverse relationship between exponentials and logarithms
- Graphs and asymptotic (long-run) behavior

Learning objectives:

- Explain the inverse relationship between exponential and logarithmic functions.
- Apply logarithm laws to simplify expressions and solve equations.
- Evaluate real-world quantities (e.g., pH, population size) using exponential/log models.

1.3 Mathematical Modeling in Bio/Social Science Contexts

Core concepts:

- Choosing a function family for a phenomenon (linear vs. exponential vs. logistic growth)
- Interpreting model parameters and units
- Qualitative model fit to data

Learning objectives:

- Explain why phenomena such as population growth or compound interest are modeled exponentially rather than linearly.
- Design a simple function-based model for a given biological or social scenario.
- Evaluate the reasonableness of a proposed model against the qualitative features of data.

Example Questions

1. (*Foundational*) A culture starts with 200 cells and doubles every 3 hours. Write a function $N(t)$ for the population after t hours.
Answer/Rubric: $N(t) = 200 \cdot 2^{t/3}$.
2. (*Applied*) A town's population was 12,000 in 2010 and 15,000 in 2020. Assuming exponential growth, find the annual growth rate and predict the 2030 population.
Answer/Rubric: Full credit requires (a) setting up $P(t) = P_0(1+r)^t$, (b) solving for r from the two data points, (c) projecting correctly to 2030, (d) a sentence noting that the exponential-growth assumption may not hold indefinitely.

2 Module 2: Limits and Continuity

Goal: Build the concept of a limit as the foundation for the derivative, and connect continuity to real-world *no sudden jumps* behavior.

2.1 Intuitive Notion of a Limit; One-Sided Limits

Core concepts:

- Limits as “approaching” behavior
- Left-hand and right-hand limits
- When a limit fails to exist
- Estimating limits from graphs and tables

Learning objectives:

- Explain the difference between a function’s value at a point and its limit at that point.
- Apply graphical and tabular reasoning to estimate one-sided and two-sided limits.

2.2 Limit Laws and Algebraic Computation

Core concepts:

- Sum, product, and quotient laws for limits
- Limits of polynomial and rational functions
- The indeterminate form $0/0$
- Factoring and rationalizing techniques

Learning objectives:

- Apply limit laws to compute limits of algebraic combinations of functions.
- Derive simplified forms of indeterminate expressions using factoring or rationalization.

2.3 Continuity and Its Role in Modeling

Core concepts:

- Definition of continuity at a point
- Types of discontinuity: removable, jump, infinite
- The Intermediate Value Theorem
- Continuity as a modeling assumption

Learning objectives:

- Explain why continuity is a natural (or sometimes unrealistic) assumption in biological/social models.
- Apply the Intermediate Value Theorem to justify the existence of a solution in context (e.g., a threshold dose, an equilibrium price).

Example Questions

1. (*Foundational*) Use a table of values to estimate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

Answer/Rubric: Values approach 4 from both sides; algebraically, $\frac{x^2 - 4}{x - 2} = x + 2 \rightarrow 4$.

2. (*Applied*) Body temperature $T(t)$ during an infection is modeled by a piecewise function that jumps when medication is administered at $t = 6$ hours. Explain, using the definition of continuity, why T is discontinuous at $t = 6$ and what this means physically.

Answer/Rubric: Strong answers note $\lim_{t \rightarrow 6^-} T(t) \neq \lim_{t \rightarrow 6^+} T(t)$, identify this as a jump discontinuity, and connect it to the instantaneous effect of the medication.

3 Module 3: The Derivative: Definition and Rates of Change

Goal: Introduce the derivative as a formalization of instantaneous rate of change, motivated by biological and social rate questions.

3.1 Average vs. Instantaneous Rate of Change

Core concepts:

- Average rate of change as the slope of a secant line
- Instantaneous rate as a limit of average rates
- Motivating examples: population growth rate, marginal cost

Learning objectives:

- Explain the conceptual jump from average to instantaneous rate of change.
- Apply the difference quotient to estimate an instantaneous rate from data.

3.2 The Derivative as a Limit; the Difference Quotient

Core concepts:

- Formal definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Computing derivatives from the definition for simple functions
- Notation: $f'(x)$, $\frac{dy}{dx}$

Learning objectives:

- Derive the derivative of simple functions (e.g., $f(x) = x^2$) directly from the limit definition.
- Apply correct derivative notation in context.

3.3 The Derivative as a Function; Differentiability and Continuity

Core concepts:

- The derivative as a new function built from f
- Graphical relationship between f and f'
- Differentiability implies continuity (not conversely)

- Points of non-differentiability: corners, cusps, vertical tangents

Learning objectives:

- Explain why differentiability implies continuity but not vice versa.
- Evaluate a graph of f to sketch a plausible graph of f' .

Difficulty Spotlight

- **The derivative as a limit (Topic 3.2):** *Why it's hard:* students face an abstraction jump from a concrete average rate to a limiting process, plus algebra-heavy simplification of the difference quotient before the limit can be taken. *Mitigation:* scaffold with a numerical table of average rates over shrinking intervals before introducing the symbolic limit; sequence worked examples from simple polynomials to slightly more complex ones.

Example Questions

1. (*Foundational*) Use the limit definition to find $f'(x)$ for $f(x) = x^2 + 3x$.

Answer/Rubric: $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} = 2x + 3$.

2. (*Applied*) A tumor's volume (in mm^3) is measured at $t = 0, 1, 2, 3$ weeks as 50, 58, 68, 80. Estimate the instantaneous growth rate at $t = 2$ using the data, and explain why this is only an estimate of the true derivative.

Answer/Rubric: Strong answers use a centered difference $\frac{80-58}{2} = 11 \text{ mm}^3/\text{week}$ as an approximation, and explain that only discrete data (not a continuous function) is available, so the true instantaneous rate cannot be computed exactly.

4 Module 4: Differentiation Rules

Goal: Equip students with the mechanical toolkit for differentiating combinations of functions. (*Quiz 1 at the end of this module.*)

4.1 Power, Sum/Difference, and Constant Multiple Rules

Core concepts:

- Power rule for x^n
- Linearity of differentiation
- Differentiating polynomials

Learning objectives:

- Apply the power rule and linearity to differentiate polynomial functions quickly.

4.2 Product Rule and Quotient Rule

Core concepts:

- Statement and use of the product rule
- Statement and use of the quotient rule
- Common errors (e.g., treating $\frac{d}{dx}(fg)$ as $f'g'$)

Learning objectives:

- Derive the product rule informally from the limit definition.
- Apply the product and quotient rules to differentiate combined functions.

4.3 Chain Rule

Core concepts:

- Composite functions
- Statement of the chain rule
- Recognizing “outside/inside” structure
- Chain rule with multiple nested layers

Learning objectives:

- Explain the chain rule as tracking how a composite function’s rate of change compounds.
- Apply the chain rule to differentiate nested/composite functions, including modeling contexts (e.g., the rate of change of a population expressed as a function of time through an intermediate variable).

Difficulty Spotlight

- **Product, quotient, and chain rules (Topics 4.2–4.3):** *Why it’s hard:* there is notational overload across three new rules in one week, and the chain rule’s “outside/inside” structure is easy to misidentify, leading to systematic sign and structure errors. *Mitigation:* use a consistent color-coded or boxed “outside/inside” annotation method for every chain-rule example, and run explicit common-error walkthroughs comparing a wrong application side by side with the correct one.

Example Questions

1. (*Foundational*) Differentiate $f(x) = 3x^4 - 5x^2 + 7$.
Answer/Rubric: $f'(x) = 12x^3 - 10x$.
2. (*Applied*) The concentration of a drug in the bloodstream is $C(t) = \frac{5t}{t^2 + 4}$ (mg/L). Find $C'(t)$ and interpret its sign at $t = 1$ and $t = 3$.
Answer/Rubric: Requires correct quotient-rule application; strong answers state that $C'(1) > 0$ (concentration still rising) and $C'(3) < 0$ (concentration falling), tying the sign of the derivative to the physical process.

5 Module 5: Derivatives of Exponential and Logarithmic Functions

Goal: Extend the differentiation toolkit to exponential and logarithmic functions, central to growth and decay modeling in the biological and social sciences.

5.1 Derivative of e^x and General Exponential Functions

Core concepts:

- $\frac{d}{dx}e^x = e^x$
- Derivative of a^x via $a^x = e^{x \ln a}$
- Combining with the chain rule for e^{kt}

Learning objectives:

- Derive the derivative of a^x from the derivative of e^x .
- Apply exponential differentiation to growth and decay rate problems.

5.2 Derivative of $\ln x$ and General Logarithmic Functions

Core concepts:

- $\frac{d}{dx} \ln x = \frac{1}{x}$
- Derivative of $\log_a x$
- Chain rule with logarithms

Learning objectives:

- Apply logarithmic differentiation rules to functions involving $\ln$ and $\log_a$.

5.3 Logarithmic Differentiation and Growth-Rate Applications

Core concepts:

- Taking $\ln$ of both sides to differentiate products, quotients, and powers
- Relative (per-capita) rate of change, f'/f
- Connection to per-capita growth rate in biology and economics

Learning objectives:

- Apply logarithmic differentiation to simplify differentiation of complex products or powers.
- Evaluate a population or economic growth scenario in terms of its relative (per-capita) growth rate.

Difficulty Spotlight

- **Logarithmic differentiation and relative rate of change (Topic 5.3):** *Why it's hard:* the topic layers a new algebraic technique on top of an unfamiliar interpretive idea (percentage rate vs. absolute rate), which is conceptually dense for one sitting. *Mitigation:* work one running example (e.g., a bacterial population) throughout the lecture, computing both f' and f'/f at the same points so students see the two numbers side by side and connect f'/f to “percent growth per unit time.”

Example Questions

1. (*Foundational*) Find $\frac{d}{dx} (e^{3x} + \ln(5x))$.

Answer/Rubric: $3e^{3x} + \frac{1}{x}$.

2. (*Applied*) A population grows as $P(t) = 500e^{0.04t}$. Compute the relative growth rate $P'(t)/P(t)$ and explain what it means that this quantity is constant.

Answer/Rubric: $P'(t)/P(t) = 0.04$ for all t ; strong answers explain that a constant relative growth rate of 4% per unit time is the defining feature of exponential growth.

6 Module 6: Derivatives of Trigonometric Functions and Implicit Differentiation

Goal: Round out the differentiation toolkit with light trigonometric derivatives and implicit differentiation for relationships not solved explicitly for y .

6.1 Derivatives of Sine and Cosine

Core concepts:

- $\frac{d}{dx} \sin x = \cos x$, $\frac{d}{dx} \cos x = -\sin x$ (statement and geometric intuition)
- Periodic behavior in rates (e.g., seasonal or circadian models)

Learning objectives:

- Apply derivatives of sine and cosine, together with the chain rule, in periodic modeling contexts (e.g., seasonal population fluctuation).

6.2 Implicit Differentiation

Core concepts:

- Differentiating equations not solved for y
- Treating y as a function of x
- Solving for $\frac{dy}{dx}$

Learning objectives:

- Apply implicit differentiation to find dy/dx for equations relating two variables implicitly.

6.3 Applications of Implicit Differentiation

Core concepts:

- Implicitly defined relationships in bio/social models (e.g., budget constraints, resource trade-off curves)
- Interpreting dy/dx in context

Learning objectives:

- Evaluate the meaning of a dy/dx obtained implicitly in a real-world trade-off scenario.

Example Questions

1. (*Foundational*) Find $\frac{dy}{dx}$ if $x^2 + y^2 = 25$.

Answer/Rubric: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -x/y$.

2. (*Applied*) Two species share a habitat with a resource trade-off curve $x^2 + 4y^2 = 400$, where x and y are population sizes. Find dy/dx when $x = 12, y = 8$ and interpret its sign.

Answer/Rubric: $dy/dx = -x/(4y)$; at $(12, 8)$, $dy/dx = -3/8$, meaning population y decreases as x increases along the trade-off curve.

7 Module 7: Related Rates

Goal: Apply the chain rule to problems where multiple quantities change with respect to time simultaneously. (*Quiz 2 at the end of this module; Group Project 1 assigned.*)

7.1 Setting Up Related-Rates Problems

Core concepts:

- Identifying variables and their rates
- Writing an equation relating the variables
- Differentiating implicitly with respect to time
- Solving for the unknown rate

Learning objectives:

- Design a solution strategy (draw, label, relate, differentiate, solve) for a related-rates problem.
- Apply the chain rule to relate the rates of two or more quantities.

7.2 Related Rates in Biological and Social Contexts

Core concepts:

- Spread of a disease relative to contact rate
- Changing population density in an expanding habitat
- Rate of change of an economic quantity relative to a driver variable

Learning objectives:

- Apply related-rates methodology to a novel biological or social scenario.
- Evaluate the reasonableness of a computed rate against the scenario's real-world constraints.

Difficulty Spotlight

- **Related rates (Topics 7.1–7.2):** *Why it's hard:* translating prose into an equation relating multiple time-dependent quantities is a classic sticking point, and students often confuse a fixed quantity with a rate. *Mitigation:* require a consistent five-step template (draw, label with rates, relate, differentiate, solve) on every problem, and progress deliberately from classic geometric setups to scenario-based ones only after the template is automatic.

Example Questions

1. (*Foundational*) A circular colony of bacteria has radius growing at 2 mm/hr. How fast is the area growing when the radius is 10 mm?
Answer/Rubric: $A = \pi r^2 \Rightarrow dA/dt = 2\pi r dr/dt = 2\pi(10)(2) = 40\pi$ mm²/hr.
2. (*Applied*) In a city, the number of infected individuals I and the number of susceptible individuals S satisfy $I \cdot S = 10,000,000$ (a simplified contact-based relationship). If S is decreasing at 500 people/day when $S = 20,000$, how fast is I changing?
Answer/Rubric: Differentiate implicitly: $I'S + IS' = 0$; at $S = 20,000$, $I = 500$; solve $I' = -IS'/S = -(500)(-500)/20,000 = 12.5$ people/day increase.

8 Module 8: Optimization

Goal: Use derivatives to find and justify optimal values in applied problems relevant to biology and social science.

8.1 Critical Points and the First/Second Derivative Tests

Core concepts:

- Critical points: $f' = 0$ or undefined
- First derivative test
- Second derivative test
- Global vs. local extrema; the Extreme Value Theorem on closed intervals

Learning objectives:

- Apply the first and second derivative tests to classify critical points.
- Evaluate whether a critical point is a global extremum on a given domain.

8.2 Optimization in Biological and Economic Contexts

Core concepts:

- Setting up an objective function from a word problem
- Identifying constraints
- Examples: maximizing yield, minimizing cost, optimal foraging/resource allocation, maximizing revenue or profit

Learning objectives:

- Design an objective function and constraint(s) from a described optimization scenario.
- Apply calculus-based optimization to solve for and justify an optimal value in context.

Difficulty Spotlight

- **Optimization word problems (Topic 8.2):** *Why it's hard:* students struggle to translate prose into a correct objective function and constraint, and often assume a critical point is automatically the desired maximum or minimum without justification. *Mitigation:* use an explicit objective/constraint/domain worksheet scaffold for every problem, and make justification via the first or second derivative test a required habit, not an afterthought.

Example Questions

1. (*Foundational*) Find the critical points of $f(x) = x^3 - 6x^2 + 9x$ and classify each using the second derivative test.
Answer/Rubric: $f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$; critical points at $x = 1, 3$. $f''(x) = 6x - 12$; $f''(1) < 0$ (local max), $f''(3) > 0$ (local min).
2. (*Applied*) A conservation program has a budget to fence a rectangular reserve using a river as one side (no fencing needed there). With 800 m of fencing available, find the dimensions that maximize the enclosed area.
Answer/Rubric: Let x = side perpendicular to river, y = side parallel. Constraint: $2x + y = 800$. Objective: $A = xy = x(800 - 2x)$. $A'(x) = 800 - 4x = 0 \Rightarrow x = 200$, $y = 400$; second derivative test confirms a maximum.

9 Module 9: Curve Sketching

Goal: Synthesize first- and second-derivative information into a complete qualitative picture of a function's graph. (*Quiz 3 at the end of this module; Portfolio 1 due.*)

9.1 Increasing/Decreasing Behavior and the First Derivative

Core concepts:

- Sign of f' and monotonicity
- Constructing sign charts
- Relating to rate-of-change interpretation (e.g., population increasing vs. decreasing)

Learning objectives:

- Apply sign analysis of f' to determine intervals of increase and decrease.

9.2 Concavity and Inflection Points

Core concepts:

- Sign of f'' and concavity
- Inflection points
- Relating concavity to accelerating/decelerating change (e.g., accelerating spread of a disease vs. growth slowing as it saturates)

Learning objectives:

- Apply sign analysis of f'' to identify concavity and inflection points.
- Explain the real-world meaning of an inflection point in a growth-curve context.

9.3 Full Curve Sketching

Core concepts:

- Combining domain, intercepts, asymptotes, and first/second-derivative information into a full sketch

Learning objectives:

- Design a complete, labeled sketch of a function using derivative information.
- Evaluate a given sketch or graph for consistency with derivative data.

Difficulty Spotlight

- **Concavity and inflection points (Topic 9.2):** *Why it's hard:* the “derivative of a derivative” idea creates cognitive overload, and students frequently confuse a function that is decreasing but concave up (or similar combinations) with contradictory scenarios. *Mitigation:* display paired graphs of f , f' , and f'' side by side for every example, and use a velocity/acceleration or “growth rate is slowing but still positive” analogy to anchor the concept.

Example Questions

1. (*Foundational*) For $f(x) = x^3 - 3x^2$, find the intervals of concavity and any inflection points.
Answer/Rubric: $f''(x) = 6x - 6$; concave down for $x < 1$, concave up for $x > 1$; inflection point at $x = 1$.
2. (*Applied*) The cumulative number of reported cases in an outbreak is modeled by $C(t)$, and public health officials note that $C''(t)$ changed sign at week 6. Explain what this means for the outbreak and why week 6 might be an important turning point.
Answer/Rubric: Strong answers identify week 6 as an inflection point: the case count is still increasing, but its rate of increase begins to slow, suggesting the outbreak is past its fastest-growth phase.

10 Module 10: Exponential Growth and Decay Models

Goal: Connect derivative concepts to simple differential-equation models of growth and decay central to biology and social science. (*Group Project 1 due; Group Project 2 assigned.*)

10.1 Modeling with $dy/dt = ky$

Core concepts:

- The differential equation $dy/dt = ky$ and its solution $y = y_0e^{kt}$
- Sign of k : growth vs. decay
- Verifying a solution by differentiation

Learning objectives:

- Explain why $dy/dt = ky$ implies exponential behavior.
- Apply the exponential solution formula to a modeling scenario given an initial condition.

10.2 Half-Life, Doubling Time, and Applications

Core concepts:

- Doubling-time and half-life formulas
- Applications: population growth, drug elimination (pharmacokinetics), radioactive decay, compound interest

Learning objectives:

- Derive the doubling-time/half-life formula from the general exponential solution.
- Apply half-life or doubling-time reasoning to a biological or financial scenario.

Example Questions

1. (*Foundational*) A radioactive isotope has a half-life of 8 days. If 100 mg are present initially, how much remains after 24 days?
Answer/Rubric: 24 days = 3 half-lives, so $100 \cdot (1/2)^3 = 12.5$ mg.
2. (*Applied*) A drug is eliminated from the bloodstream following $dC/dt = -0.2C$ (C in mg/L, t in hours). If the initial concentration is 50 mg/L, find the concentration after 5 hours and the time until the concentration drops below 5 mg/L.
Answer/Rubric: $C(t) = 50e^{-0.2t}$; $C(5) = 50e^{-1} \approx 18.4$ mg/L. Solve $50e^{-0.2t} = 5 \Rightarrow t = 5 \ln(10) \approx 11.5$ hours.

11 Module 11: Antiderivatives and Indefinite Integrals

Goal: Introduce the reverse process of differentiation and its role in reconstructing quantities from known rates.

11.1 Antiderivatives and Basic Integration Rules

Core concepts:

- Antiderivative as “reverse derivative”
- Power rule for integration
- The constant of integration
- Basic rules for exponential functions

Learning objectives:

- Apply basic antidifferentiation rules to find general antiderivatives.

11.2 Initial Value Problems and Reconstructing Functions from Rates

Core concepts:

- Using an initial condition to solve for the constant of integration
- Reconstructing a quantity (e.g., population, total cost) from its rate function

Learning objectives:

- Apply an initial condition to determine a particular antiderivative.
- Design a solution to a “reconstruct the total from the rate” word problem (e.g., total drug amount from an infusion rate).

Example Questions

1. (*Foundational*) Find the general antiderivative of $f(x) = 4x^3 - 2x + 1$.

Answer/Rubric: $F(x) = x^4 - x^2 + x + C$.

2. (*Applied*) A patient receives a drug at a rate of $r(t) = 10e^{-0.1t}$ mg/hr. If no drug is present at $t = 0$, find the total amount of drug administered by time t , $A(t)$.

Answer/Rubric: $A(t) = \int_0^t 10e^{-0.1s} ds = 100(1 - e^{-0.1t})$ mg, using $A(0) = 0$ to fix the constant.

12 Module 12: The Definite Integral and the Fundamental Theorem of Calculus

Goal: Formalize accumulated change via Riemann sums and connect it to antidifferentiation through the Fundamental Theorem of Calculus. (*Quiz 4 at the end of this module.*)

12.1 Riemann Sums and the Definite Integral

Core concepts:

- Partitioning an interval
- Left, right, and midpoint Riemann sums
- The definite integral as a limit of Riemann sums
- Geometric interpretation as (signed) area

Learning objectives:

- Explain the definite integral as a limit of Riemann sums approximating accumulated change.
- Apply Riemann sum approximations to estimate a definite integral from data (e.g., estimating total drug exposure from concentration readings).

12.2 The Fundamental Theorem of Calculus

Core concepts:

- FTC Part 1: the derivative of an accumulation function
- FTC Part 2: evaluating definite integrals via antiderivatives
- Using the FTC to compute total change

Learning objectives:

- Explain how the Fundamental Theorem of Calculus links differentiation and integration.
- Apply FTC Part 2 to evaluate definite integrals and compute total change from a rate function.

Difficulty Spotlight

- **Riemann sums and the definite integral (Topic 12.1):** *Why it's hard:* the limiting process is abstract, sigma notation adds notational overload, and the connection between “area” and non-geometric accumulation (e.g., total births) is not obvious to students. *Mitigation:* build up from concrete numerical left/right sum tables using a real rate table (e.g., births per month) before introducing sigma notation or the symbolic limit definition, so students accumulate an actual number before seeing the formalism.

Example Questions

1. (*Foundational*) Evaluate $\int_1^3 (2x + 1) dx$ using the Fundamental Theorem of Calculus.

Answer/Rubric: Antiderivative $x^2 + x$; $(9 + 3) - (1 + 1) = 10$.

2. (*Applied*) A river's flow rate is recorded (in m^3/s) at 4 evenly spaced times over a 12-hour period: 20, 35, 50, 40. Use a Riemann sum (with these four values, each representing a 3-hour interval) to estimate total water volume that passed in 12 hours.

Answer/Rubric: Using a left Riemann sum: $(20 + 35 + 50 + 40) \times 3 \text{ hr} \times 3600 \text{ s/hr}$; strong answers correctly convert hours to seconds and note this is only an approximation of the true accumulated volume.

13 Module 13: Applications of Integration

Goal: Apply the definite integral to compute areas, averages, and accumulated change in applied bio/social contexts.

13.1 Area Between Curves

Core concepts:

- Setting up integrals for the area between two curves
- Identifying which curve is “on top”
- Applications: comparing two growth trajectories

Learning objectives:

- Apply integration to compute the area between two curves in a modeling context.

13.2 Average Value of a Function

Core concepts:

- Average value formula $\frac{1}{b-a} \int_a^b f(x) dx$
- The Mean Value Theorem for Integrals
- Interpreting average value (e.g., average concentration, average population over a period)

Learning objectives:

- Apply the average value formula to compute a meaningful average from a rate or quantity function.

13.3 Accumulated Change in Applied Contexts

Core concepts:

- Total change as $\int(\text{rate}) dt$
- The net change theorem
- Examples: total births minus deaths, cumulative economic output, total drug cleared

Learning objectives:

- Apply the net change theorem to compute total accumulated change from a rate function in a biological or social scenario.
- Evaluate whether a computed accumulated change is consistent with the underlying scenario's constraints.

Example Questions

1. (*Foundational*) Find the area between $f(x) = x^2$ and $g(x) = 2x$ on $[0, 2]$.

Answer/Rubric: $\int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$.

2. (*Applied*) Two competing species' growth rates (individuals/year) over a 5-year study are modeled by $r_1(t) = 100 - 10t$ and $r_2(t) = 60$. Find the total difference in population growth between the two species over the 5 years, and interpret which species grew more overall.

Answer/Rubric: $\int_0^5 [r_1(t) - r_2(t)] dt = \int_0^5 (40 - 10t) dt = [40t - 5t^2]_0^5 = 200 - 125 = 75$; species 1 grew 75 more individuals overall despite its declining rate.

14 Module 14: Differential Equations: Separable Equations and Logistic Growth

Goal: Extend growth modeling beyond pure exponential growth to more realistic, resource-limited (logistic) growth using separable differential equations. (*Portfolio 2 and Group Project 2 due.*)

14.1 Separable Differential Equations

Core concepts:

- Identifying a separable ODE
- Separating variables and integrating both sides
- Solving for the general and particular solution

Learning objectives:

- Apply separation of variables to solve a first-order separable differential equation.

14.2 The Logistic Growth Model

Core concepts:

- The logistic ODE $\frac{dy}{dt} = ky \left(1 - \frac{y}{M}\right)$
- Carrying capacity M
- Qualitative behavior: the S-shaped curve
- Comparison to unconstrained exponential growth

Learning objectives:

- Explain why logistic growth is a more realistic model than pure exponential growth for populations with limited resources.
- Evaluate the long-term behavior of a logistic model given specific parameters (k, M, y_0) .

Difficulty Spotlight

- **The logistic growth model (Topic 14.2):** *Why it's hard:* students tend to over-apply exponential-decay intuition, and the idea of a carrying capacity as a horizontal asymptote approached from below is not immediately intuitive from the differential equation alone. *Mitigation:* before writing the ODE, have students graph per-capita growth rate (y'/y) versus population size y as a downward-sloping line, then overlay the resulting logistic curve directly against an exponential curve on the same axes so the flattening near M is visually explicit.

Example Questions

1. (*Foundational*) Solve the separable equation $\frac{dy}{dx} = 3y$, $y(0) = 2$.

Answer/Rubric: Separate: $\int \frac{dy}{y} = \int 3 dx \Rightarrow \ln |y| = 3x + C \Rightarrow y = 2e^{3x}$.

2. (*Applied*) A fish population in a lake follows logistic growth with carrying capacity $M = 5000$, growth constant $k = 0.3$, and initial population $y_0 = 500$. Describe qualitatively how the population changes over time, including where growth is fastest.

Answer/Rubric: Strong answers describe an S-shaped increase from 500 toward 5000, note the population never exceeds M , and identify that growth is fastest near $y = M/2 = 2500$ (the inflection point of the logistic curve).

Week 15: Review and Final Exam

The final week is reserved for cumulative review (synthesizing differentiation, integration, and modeling applications across all 14 modules) followed by the final exam, which counts for 30% of the course grade.

15 Prerequisite Map

Topic	Prerequisite(s)	Where Taught
1.1 Function Types Review	College algebra: function notation, graphing basic function families	Assumed background
1.2 Exponential/Log Functions	1.1; precalculus exponent and logarithm rules	Module 1; Assumed background
1.3 Modeling	1.1, 1.2	Module 1

Topic	Prerequisite(s)	Where Taught
2.1 Intuitive Limits	1.1 (graphs of functions)	Module 1
2.2 Limit Laws	2.1; algebraic factoring/rationalizing	Module 2; Assumed background
2.3 Continuity	2.1, 2.2	Module 2
3.1 Average vs. Instantaneous Rate	1.1 (slope of a line); 2.1 (limit intuition)	Module 1; Module 2
3.2 Derivative as a Limit	2.2 (limit laws); 3.1	Module 2; Module 3
3.3 Derivative as a Function	3.2	Module 3
4.1 Power/Sum/Constant Rules	3.2 (derivative definition); precalculus exponent rules	Module 3; Assumed background
4.2 Product/Quotient Rules	4.1	Module 4
4.3 Chain Rule	4.1, 4.2; function composition	Module 4; Assumed background
5.1 Derivative of e^x/a^x	4.3 (chain rule); 1.2 (exponential functions)	Module 4; Module 1
5.2 Derivative of $\ln x/\log_a x$	5.1; 1.2 (logarithmic functions)	Module 5; Module 1
5.3 Logarithmic Differentiation	5.1, 5.2	Module 5
6.1 Derivatives of Sine/Cosine	4.3 (chain rule); precalculus trigonometric functions	Module 4; Assumed background
6.2 Implicit Differentiation	4.3 (chain rule)	Module 4
6.3 Applications of Implicit Diff.	6.2	Module 6
7.1 Setting Up Related Rates	4.3 (chain rule); 6.2 (implicit differentiation)	Module 4; Module 6
7.2 Related Rates in Context	7.1	Module 7
8.1 Critical Points & Derivative Tests	3.3 (derivative as a function)	Module 3
8.2 Optimization in Context	8.1; 1.3 (modeling)	Module 8; Module 1
9.1 Increasing/Decreasing	8.1	Module 8
9.2 Concavity & Inflection Points	9.1 (second-derivative concept builds on first)	Module 9
9.3 Full Curve Sketching	9.1, 9.2; 1.1 (graphing basics)	Module 9; Module 1
10.1 Modeling with $dy/dt = ky$	5.1 (derivative of exponential); 4.1	Module 5; Module 4
10.2 Half-Life/Doubling Time	10.1	Module 10
11.1 Antiderivatives	4.1 (power rule, reversed); 5.1 (exponential derivative, reversed)	Module 4; Module 5
11.2 Initial Value Problems	11.1	Module 11
12.1 Riemann Sums	1.1 (area/graph intuition); summation notation	Module 1; Assumed background
12.2 Fundamental Theorem of Calculus	12.1; 11.1 (antiderivatives)	Module 12; Module 11
13.1 Area Between Curves	12.2 (FTC); 1.1 (comparing graphs)	Module 12; Module 1
13.2 Average Value	12.2 (FTC)	Module 12
13.3 Accumulated Change	12.2 (FTC)	Module 12
14.1 Separable Differential Equations	11.1 (antiderivatives); algebraic manipulation	Module 11; Assumed background
14.2 Logistic Growth	14.1; 10.1 (exponential growth, for contrast)	Module 14; Module 10

16 Assessment Plan

Module	Assessment & Timing	Rationale
Module 1	Homework 1 (ongoing weekly)	Diagnostic, low-stakes check of function fluency and modeling intuition before calculus begins.
Module 2	Homework 2	Reinforces limit computation and continuity reasoning while still low-stakes.
Module 3	Homework 3	Practice with the formal derivative definition, the conceptual core of the course.
Module 4	Homework 4; Quiz 1	Quiz checks mechanical fluency with differentiation rules before students move on to applications.
Module 5	Homework 5	Practice differentiating exponential/log functions, essential for the growth-model unit ahead.
Module 6	Homework 6	Practice with trig derivatives and implicit differentiation before related rates requires both.
Module 7	Homework 7; Quiz 2; Group Project 1 assigned	Quiz checks related-rates setup skill; the project lets students apply derivative concepts to an open-ended bio/social scenario over several weeks.
Module 8	Homework 8	Practice with optimization setups, building toward the curve-sketching synthesis in Module 9.
Module 9	Homework 9; Quiz 3; Portfolio 1 due	Quiz checks curve-sketching synthesis; the portfolio consolidates reflection on the full differentiation unit (Modules 3–9) at the course’s structural midpoint.
Module 10	Homework 10; Group Project 1 due; Group Project 2 assigned	Marks the transition from differentiation to growth modeling and integration; project cycle continues into the second half.
Module 11	Homework 11	Practice with antidifferentiation, the foundation for the definite integral.
Module 12	Homework 12; Quiz 4	Final quiz checks fluency with Riemann sums and the Fundamental Theorem before the applications unit.
Module 13	Homework 13	Practice applying integration to area, average value, and accumulated change.
Module 14	Homework 14; Portfolio 2 due; Group Project 2 due	Final portfolio and project consolidate the full course and are submitted before cumulative review.
Week 15	Final Exam	Cumulative assessment covering differentiation, integration, and modeling applications across the whole course.